Applied Numerical Methods for Civil Engineering

CGN 3405 - 0002

Week 13: Optimization techniques: Part II

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Linear & Quadratic Programming

More material:

- Linear programming slides:

  https://xinychen.github.io/youtube/linprog.pdf

- Linear programming explained:

  https://www.youtube.com/watch?v=Wt8oSh26BY8

Derivatives of $f(x)$ and $f(x, y)$

A quick revisit!

- **Derivative.** Given a scalar function $f(x)$ of the single variable x, the derivative is defined by

$$\frac{\mathrm{d}\, f(x)}{\mathrm{d}\, x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x}$$

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- **Partial derivatives.** Given a scalar function $f(x, y)$ of two variables x, y, the partial derivatives are defined by

$$\begin{cases} \dfrac{\partial f(x, y)}{\partial x} = \lim\limits_{\Delta x \to 0} \dfrac{f(x + \Delta x, y)}{\Delta x} \\ \dfrac{\partial f(x, y)}{\partial y} = \lim\limits_{\Delta y \to 0} \dfrac{f(x, y + \Delta y)}{\Delta y} \end{cases}$$

Derivative of $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_2^2$

Given $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_2^2$, write down the derivative $\dfrac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}}$:

- The function $f(\boldsymbol{x})$ can be written as

$$f(\boldsymbol{x}) = f(x_1, x_2, \ldots, x_n) = x_1^2 + x_2^2 + \cdots + x_n^2$$

- The partial derivatives of $f(x_1, x_2, \ldots, x_n)$ with respect to $x_1, x_2, \ldots, x_n$ are

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$$\begin{aligned}
\frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_1} &= 2x_1 \\
\frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_2} &= 2x_2 \\
&\vdots \\
\frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_n} &= 2x_n
\end{aligned}
\quad \Rightarrow \quad
\frac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}} = \begin{bmatrix} \frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_1} \\ \frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_n} \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \\ \vdots \\ 2x_n \end{bmatrix} = 2\boldsymbol{x}$$

Derivative of $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$

For any symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, suppose $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$, write down the derivative $\dfrac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}}$:

- The function $f(\boldsymbol{x})$ can be written as

$$f(\boldsymbol{x}) = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i A_{ij} x_j$$

- The partial derivatives of $f(\boldsymbol{x})$:

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- The partial derivatives of $f(\boldsymbol{x})$:

$$\frac{\partial f(\boldsymbol{x})}{\partial x_h} = \sum_{i=1}^{n} A_{ih} x_h + \sum_{j=1}^{n} x_h A_{hj} \quad h = 1, 2, \ldots, n$$

- If $\boldsymbol{A}$ is symmetric, we have

$$\frac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{A}^\top \boldsymbol{x} = 2\boldsymbol{A}\boldsymbol{x}$$

Derivative of $f(\boldsymbol{x}) = \boldsymbol{b}^\top \boldsymbol{x}$

For any vector $\boldsymbol{b} \in \mathbb{R}^n$, suppose $f(\boldsymbol{x}) = \boldsymbol{b}^\top \boldsymbol{x}$, write down the derivative $\dfrac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}}$:

- The function $f(\boldsymbol{x})$ can be written as

$$f(\boldsymbol{x}) = \sum_{i=1}^{n} b_i x_i$$

- The partial derivatives of $f(\boldsymbol{x})$:

Derivative of $f(\boldsymbol{x}) = \boldsymbol{b}^\top \boldsymbol{x}$

For any vector $\boldsymbol{b} \in \mathbb{R}^n$, suppose $f(\boldsymbol{x}) = \boldsymbol{b}^\top \boldsymbol{x}$, write down the derivative $\frac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}}$:

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$$\frac{\partial f(\boldsymbol{x})}{\partial x_h} = b_h \quad h = 1, 2, \ldots, n$$

- So, we have

$$\frac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}} = \boldsymbol{b}$$

Optimization over Quadratic Form

- Function of quadratic form:

$$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x} + c$$

  with symmetric and positive definite matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ and known $\boldsymbol{b} \in \mathbb{R}^n, c \in \mathbb{R}$.

- Minimizing $f(\boldsymbol{x})$:

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}$$

- Derivative of $f(\boldsymbol{x})$:

$$\frac{\mathrm{d}\, f(\boldsymbol{x})}{\mathrm{d}\, \boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}$$

- Let the derivative be zero, it gives us the optimal values:

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- Let the derivative be zero, it gives us the optimal values:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \quad \Rightarrow \quad \boldsymbol{x}^\star = \boldsymbol{A}^{-1}\boldsymbol{b}$$

Linear Systems

- A system of linear equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

- Rewrite the problem:

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}}_{\boldsymbol{A} \in \mathbb{R}^{n \times n}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{\boldsymbol{x} \in \mathbb{R}^n} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_{\boldsymbol{b} \in \mathbb{R}^n}$$

- Matrix form $\boldsymbol{Ax} = \boldsymbol{b}$

Update Equations

How to solve $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$?

- Essential idea: **Gradient descent** solves linear system iteratively.
- $\boldsymbol{x}$ is updated by

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \underbrace{\alpha}_{\text{step size}} \boldsymbol{g}_k$$

  with initial points $\boldsymbol{x}_0$ and known constant α
- $\boldsymbol{g}_k$ is the gradient:

$$\boldsymbol{g}_k = \boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b}$$

- If the residual is $\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$, then

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \underbrace{(-\alpha(\boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b}))}_{\text{step size} \times \text{negative gradient}}$$

$$\Rightarrow \quad \boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \underbrace{\alpha \boldsymbol{r}_k}_{\text{step size} \times \text{residual}}$$

Gradient Descent

- Update equations:

$$\boldsymbol{g}_k = \boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b} \qquad \text{(gradient)}$$
$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha \boldsymbol{g}_k \qquad \text{(variables)}$$

- Problem statement:

$$\begin{aligned} 4x_1 + x_2 &= 1 \\ x_1 + 3x_2 &= 2 \end{aligned} \qquad \text{with} \qquad \boldsymbol{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

  Suppose we have initial points $x_1 = 0, x_2 = 0$ (i.e., $\boldsymbol{x}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ for the initialization) and step size $\alpha = \frac{1}{4}$.

- Exact solution: $x_1 = \frac{1}{11} \approx 0.09, x_2 = \frac{7}{11} \approx 0.63$

A Simple Linear System

Given $\boldsymbol{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\boldsymbol{x}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and $\alpha = \frac{1}{4}$, we have

- **First iteration**:

$$\boldsymbol{g}_0 = \boldsymbol{A}\boldsymbol{x}_0 - \boldsymbol{b} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$\boldsymbol{x}_1 = \boldsymbol{x}_0 - \alpha \boldsymbol{g}_0 = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}$$

- **Second iteration**:

$$\boldsymbol{g}_1 = \boldsymbol{A}\boldsymbol{x}_1 - \boldsymbol{b} = \begin{bmatrix} 1/2 \\ -1/4 \end{bmatrix}$$

$$\boldsymbol{x}_2 = \boldsymbol{x}_1 - \alpha \boldsymbol{g}_1 = \begin{bmatrix} 1/8 \\ 9/16 \end{bmatrix}$$

- **Third iteration**:

$$\boldsymbol{g}_2 = \boldsymbol{A}\boldsymbol{x}_2 - \boldsymbol{b} = \begin{bmatrix} 1/16 \\ -3/16 \end{bmatrix}$$

$$\boldsymbol{x}_3 = \boldsymbol{x}_2 - \alpha \boldsymbol{g}_2 = \begin{bmatrix} 7/64 \\ 39/64 \end{bmatrix} \approx \begin{bmatrix} 0.109 \\ 0.609 \end{bmatrix}$$

Quick Summary

Today's Class:

- Derivatives of $f(\boldsymbol{x})$
- Gradient descent for minimizing

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}$$

- Solving linear systems $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$